THE LARGEST OPEN WHERE A TAU-LOCAL PROPERTY HOLDS ALONG S IS R-INVARIANT

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- tau-local properties of morphisms. Descent on Spaces, definition property-morphisms-local and lemma largest-open-of-the-base (cross-chapter, named). Provides the maximal open W and the analogue W_1 over R.
- Invariant opens for groupoids. Groupoids in Spaces, definition invariant-open (cross-chapter, named). Target notion: s^{-1}(W) = t^{-1}(W).

Source reference. Original equation 01GQ: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 284–284. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 020B: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 262–262. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03YH: focused-seeds/geometry/S9-smg-property-invariant/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06R2: focused-seeds/geometry/S9-smg-property-invariant/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 044F: focused-seeds/geometry/S9-smg-property-invariant/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $B \to S$ be as in Section 020B. Let (U, R, s, t, c) be a groupoid in algebraic spaces over B. Let $\tau \in \{fppf, étale, smooth, syntomic\}$. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces which is τ-local on the target (Descent on Spaces, Definition 03YH). Assume $\{s : R \to U\}$ and $\{t : R \to U\}$ are coverings for the τ-topology. Let $W \subset U$ be the maximal open subspace such that $s^{-1}(W) \to W$ has property $\mathcal{P}$. Then W is R-invariant (Groupoids in Spaces, Definition 044F).*

Proof. The existence and properties of the open $W \subset U$ are described in Descent on Spaces, Lemma 06R2. In Diagram (01GQ) let $W_1 \subset R$ be the maximal open subscheme over which the morphism $\mathrm{pr}_1 : R \times_{s,U,t} R \to R$ has property $\mathcal{P}$. It follows from the aforementioned Descent on Spaces, Lemma 06R2 and the assumption that $\{s : R \to U\}$ and $\{t : R \to U\}$ are coverings for the τ-topology that $t^{-1}(W) = W_1 = s^{-1}(W)$ as desired. □